

UHF RFID Reader Key Parameters Detailed Table

Parameter Name	Settings/Range	Description & Function	Impact & Configuration Strategy
Q Value	0 ~ 15 (Fixed or Dynamic)	The core parameter of the anti-collision algorithm, determining the number of inventory slots (2^Q) and regulating tag response probability. In Default Dynamic Q mode, the reader adjusts this value automatically based on the number of tags identified.	High Tag Density(>20): Increase Q value (7-15) to expand the response window, reduce collisions , and improve inventory accuracy. Low Tag Density(<5): Decrease Q value (0-4) to shorten response time and increase inventory speed . General or Changing Scenarios: Enable Dynamic Q for automatic optimization, balancing efficiency and accuracy.
Session	0 (S0) 1 (S1) 2 (S2) 3 (S3) 4 (Auto)	Controls the persistence and transition rules of the tag's "A(Responding)/B(Sleeping)" state, used to solve the repeated reading problem in multi-tag environments. Each session layer operates independently.	S0: Non-persistent state. Tag reverts to state A immediately after leaving the RF field. Suitable for single-tag or very simple scenarios. S1 (Most Common): Tag automatically reverts to state A within 500ms~5s after entering state B. Ideal for high-frequency inventory in dynamic environments (e.g., conveyor belts). S2/S3: Persistent state. Tag remains in state B for at least 2 seconds after leaving the field, effectively reducing repeated reads. Suitable for dense, static tag environments (e.g., warehouses). Auto: Switches interactively between S2 and S3 queries. Optimal for inventory in fixed, enclosed tag environments.

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TID Address	0 ~ 32 (Word Address)	Specifies the starting word address in the TID (Tag Identifier) memory bank from which to begin reading data.	Start at 0: Starts reading from the beginning of the TID bank (often contains manufacturer info). Start at other addresses (e.g., 2): Can be used to skip fixed header information and read directly to the section containing variable data or unique IDs.
TID Length	0 or Other length value (Words)	Specifies the length of TID data to read. This parameter determines whether the operation targets the EPC bank or the TID bank.	Set to 0: Does not read TID; the operation target is the EPC bank (for fast tag inventory). Set to a value >0 (e.g., 6): Reads TID data of the specified length starting from the TID Address (for authentication, distinguishing chip types).
Inventory Duration (ScanTime)	0ms ~ 25500ms (25.5 seconds)	The maximum duration for a single inventory operation . During this time, the reader continuously transmits energy and receives tag responses.	0ms: Typically signifies a single query, stopping after reading the currently polled tags. 100-500ms: Suitable for moving tags/handheld readers, requiring fast refresh rates for real-time performance. 1000-25500ms: Suitable for static inventory (e.g., warehouse checks, fixed readers), ensuring sufficient time to cover all tags.

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Interval Time	0ms ~ 2000ms (2 seconds)	The dwel (sleep) time between two consecutive Inventory Duration periods. Together with <code>ScanTime</code> , it determines the duty cycle.	0ms : No interval, continuous inventory. Highest performance , but also highest power consumption and heat generation. 10-100ms : Balances performance and power consumption, suitable for most continuous operation scenarios. >100ms : Significantly reduces power consumption, ideal for battery-powered devices , but decreases the data update rate.
Data Encoding	0 (HEX) 1 (ASCII)	Specifies the display and output format for returned data.	0 (HEX) : Returns a Hexadecimal string (e.g., <code>E200001B6600080F</code>). This is the most common, raw format, facilitating processing and transmission. 1 (ASCII) : Attempts to convert the HEX data bytes into ASCII characters . Note : Only meaningful if the stored data is actually ASCII text (e.g., "ABC123"); otherwise, it will display garbled characters.

Parameter Name	Settings/Range	Description & Function	Impact & Configuration Strategy
Power	0 ~ 33 (Unit: dBm)	The RF transmission power of the reader. Directly affects read range, coverage, and energy transfer.	Higher Value: Longer read range, larger field coverage, better penetration. However, power consumption is higher, and it may interfere with other nearby devices. Lower Value: Shorter read range, lower power consumption, more focused field strength, less likely to cause interference. Configuration Advice: Use the lowest power sufficient for the required read distance to comply with local radio regulations (e.g., the maximum power limit in China is 33dBm or 2W) and to reduce power consumption and interference.